

Amit Agarwal

AI Research Leader | Research Strategy | Enterprise Agents & Multimodal AI

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Professional Summary

Applied AI research leader with **10+ years** connecting research direction, shared AI infrastructure, and enterprise delivery. Leads research and architecture across agents, retrieval, multimodal and multilingual models, and evaluation, with an emphasis on reusable capabilities that support multiple product teams.

- **Enterprise scope:** Led agentic and knowledge systems across **15+ engagements**; delivered document AI capabilities to **20+ customers**.
- **Research impact:** First-author work on multilingual alignment and multimodal evaluation; co-first-author enterprise retrieval study achieved **0.64 MRR@10 vs. 0.57** for ADORE+STAR on a cloud-services test set.
- **Recognition and invention:** Co-author of *AccessEval*, recipient of the **EMNLP 2025 Social Impact Award**; inventor on issued and pending patents.

Experience

Oracle Cloud Infrastructure (OCI)

Apr 2021 - Present

Principal Applied Scientist (current), Generative AI

Seattle, WA

- Lead research and architecture for agentic AI and foundation-model products, connecting research priorities with Support and ERP agents, natural-language-to-SQL (NL2SQL), and code-generation systems.
- Architected a shared knowledge store for retrieval-augmented generation (RAG) and agents, bringing knowledge graphs, multimodal and multilingual retrieval, in-house rerankers, and access-aware grounding into a common platform.
- Coordinated science, engineering, and Support on pre-production evaluation readiness; verified service access and delivered onboarding. Current work extends governed context delivery through REST/MCP and retrieval evaluation harnesses.
- Led a multi-agent platform across **15+ enterprise engagements**, using tool-use evaluations and customer feedback to guide system development and integration.
- Co-owned multimodal RAG science for Oracle Fusion, contributing to an image-to-text capability deployed to production in August 2025. Led retrieval and ranking work and trained in-house embedding and ranking models.
- Delivered optical character recognition (OCR), visual question answering (VQA), layout understanding, and key information extraction (KIE) capabilities to **20+ customers**; translated document AI research into reusable product capabilities.

Jio Platforms (Reliance)

Jun 2020 - Apr 2021

Data Science Lead, Computer Vision

Hyderabad, India

- Led computer-vision development from data and models through evaluation and product integration, covering detection, tracking, scene understanding, and security/compliance for video analytics.
- Built explainable medical imaging for Reliance Hospitals and OCR-based document understanding and information retrieval.

Abzooba Infotech (acquired by UST Global)

Jun 2016 - Jun 2020

Data Scientist

Pune, India

- Served as product owner and technical lead for financial-news recommendation and shipping-disclosure classification systems; developed query understanding, named entity recognition, and implicit-feedback models.
- Built document search, summarization, and image disambiguation, along with customer-feedback analytics and predictive workflows for a Singapore government SmartCity program.

Selected Publications

- **AccessEval: Benchmarking Disability Bias in Large Language Models**
EMNLP 2025 | Social Impact Award. Co-author. Evaluated **21 LLMs across 6 domains and 9 disability types**, using paired queries to measure changes in factual accuracy, sentiment, and social perception.
- **Aligning LLMs for Multilingual Consistency in Enterprise Applications**
EMNLP 2025 Industry. First author. Batch-aligned preference optimization; Llama-3.1-70B achieved **74.9% average MGSM exact match vs. 57.2%** for the baseline in the reported study.
- **Hard Negative Mining for Domain-Specific Retrieval in Enterprise Systems**
ACL 2025 Industry. Co-first author (equal contribution). Hard-negative mining for enterprise rerankers; reported **MRR@10 of 0.64 vs. 0.57** for ADORE+STAR on a cloud-services test set.
- **Do Image-Text Metrics Respect Semantic Invariances?**
ACL 2026 Findings. First author. Evaluated five image-text metrics under meaning-preserving changes; post-hoc calibration roughly halved median sensitivity in the reported experiments.
- **RCI: A Score for Evaluating Global and Local Reasoning in Multimodal Benchmarks**
EMNLP 2025 Industry. First author. Compared full-image and image-patch performance across **13 multimodal benchmarks** to expose local shortcuts and quantify global reasoning demand.

Additional work spans ACL, EMNLP, NAACL, CVPR, ICLR, COLING, DASFAA, and CIKM. Full publication list: [Google Scholar](#).

Technical Skills

Search and agents: Retrieval-augmented generation (RAG), embeddings, reranking, knowledge graphs, multimodal and multilingual retrieval, agent orchestration, REST, Model Context Protocol (MCP).

Model development: Supervised fine-tuning (SFT), direct preference optimization (DPO), RLHF, reward modeling, distillation, quantization, synthetic data, document AI, graph neural networks.

Evaluation: Retrieval and agent evaluation harnesses, benchmark design, baseline comparisons, safety evaluation, image-text metrics.

Engineering: Python, PyTorch, TensorFlow, PyTorch Lightning, OCI, AWS, Azure, Docker, Git.

Education

MS, Machine Learning & Artificial Intelligence, Distinction, Liverpool John Moores University

BTech, Computer Science & Engineering, IEM, West Bengal University of Technology

Research Service & Open Collaboration

- **Workshop organizing:** GRAIL-V at CVPR 2026, SURGeLLM at ACL 2026, and DocInsights at EMNLP 2026.
- **Research service:** Area chair, program committee member, and reviewer across AAAI, NeurIPS, ACL, CVPR, EMNLP, NAACL, IJCAI, and ARR.
- **Mentoring and talks:** Speaker at MIPR 2024 and ICDMAI 2020; data-science mentor with Great Learning, MIT, and UT Austin programs.
- **Open source:** Contributions and collaborations across VLMEvalKit, EasyOCR, PyTorch Lightning, and Flair.

Awards

EMNLP 2025 Social Impact Award (AccessEval) | **Oracle Edison Award** | Academic Star, IIIT Bangalore | Top Performer, Abzooba